- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

**7** In this question the $x$- and $y$-directions are horizontal and vertically upwards respectively and the origin is on horizontal ground.

A ball is thrown from a point 5 m above the origin with an initial velocity $\begin{pmatrix} 14 \\ 7 \end{pmatrix} \mathrm{m\,s^{-1}}$.

**(a)** Find the position vector of the ball at time $t$ s after it is thrown. **[3]**

**(b)** Find the distance between the origin and the point at which the ball lands on the ground. **[3]**